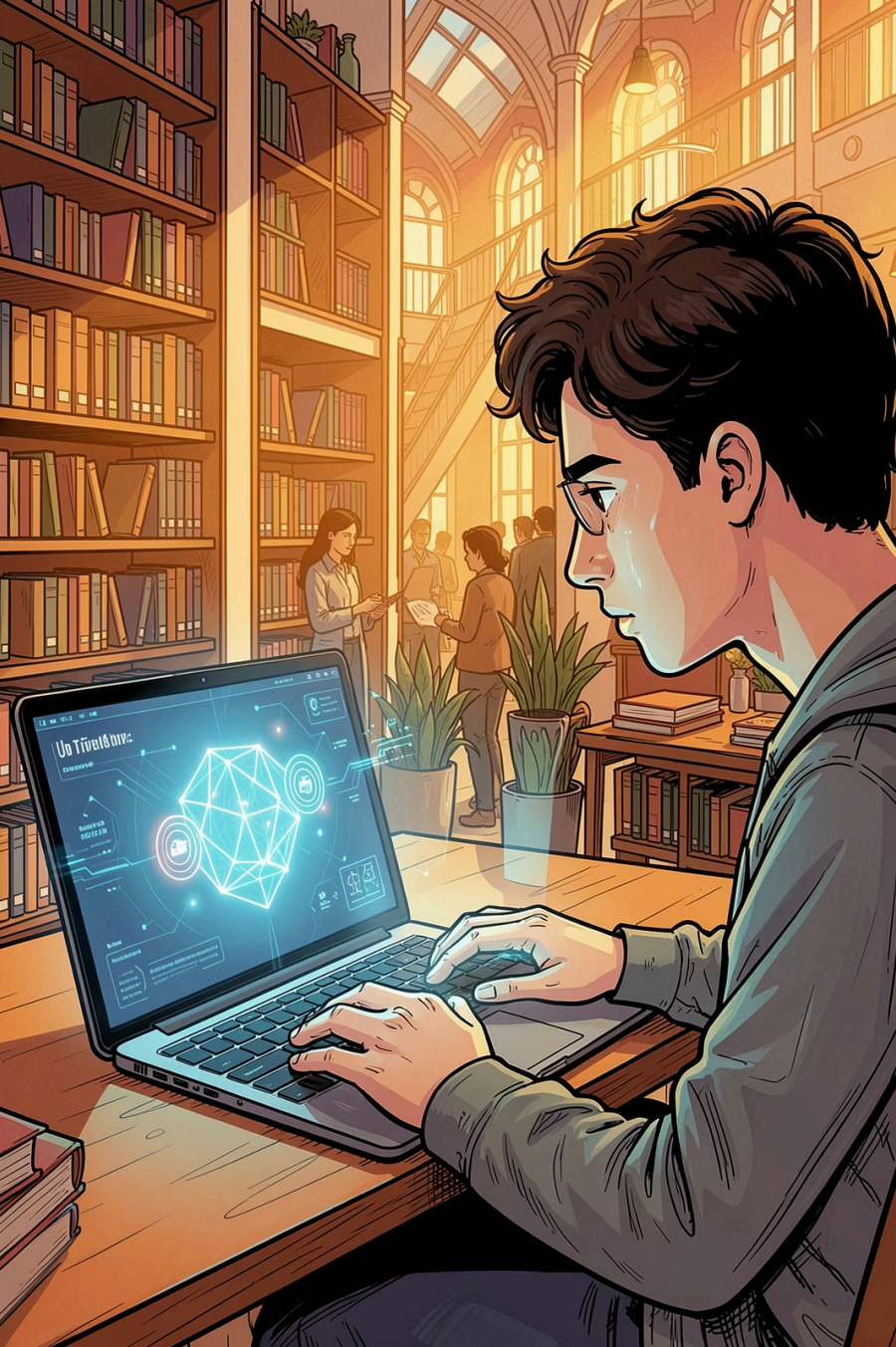


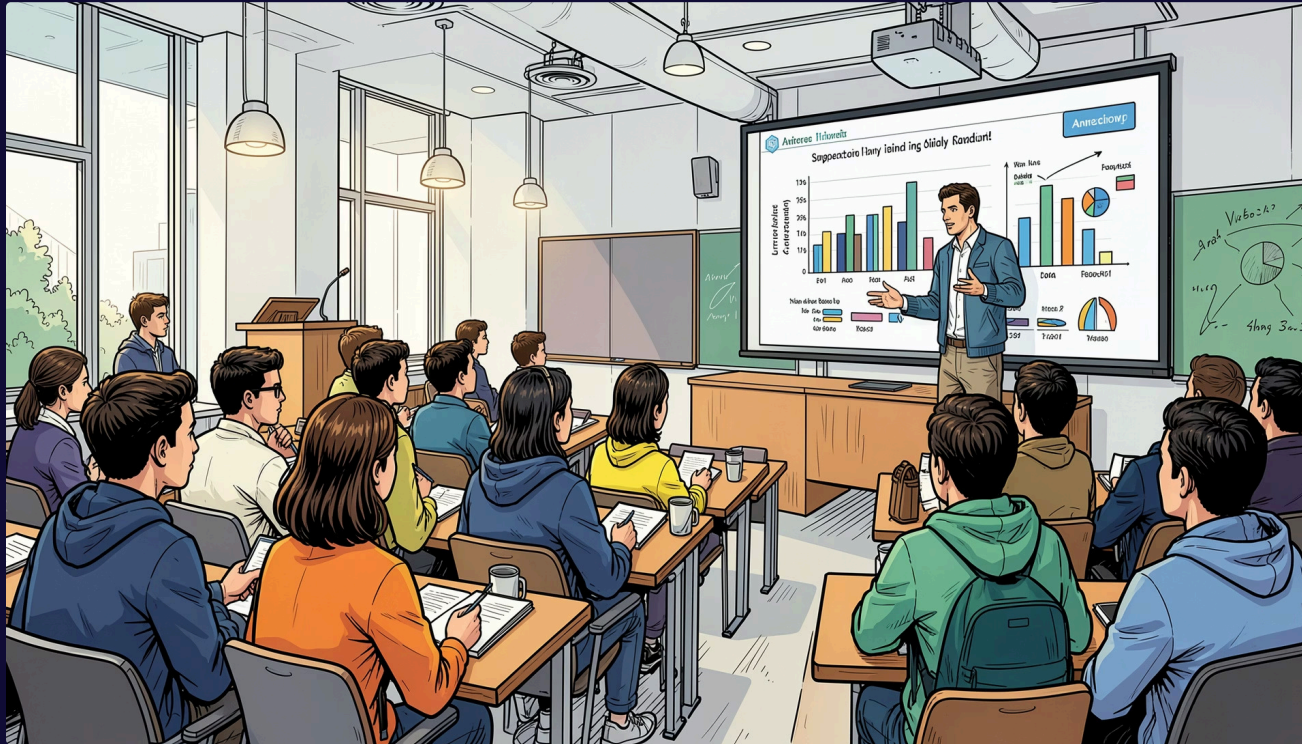


AI's Impact on Student Performance and Wellbeing

A Data Analytics Project — Analyzing how Generative AI influences academic performance, burnout, productivity, and skill retention among university students.

PREPARED BY: DEEPIKA

Overview & Problem Statement



Research Context

Generative AI is rapidly reshaping how students learn, study, and perform. This project applies data analytics to understand its real-world impact on academic outcomes and student wellbeing.

Problem Statement

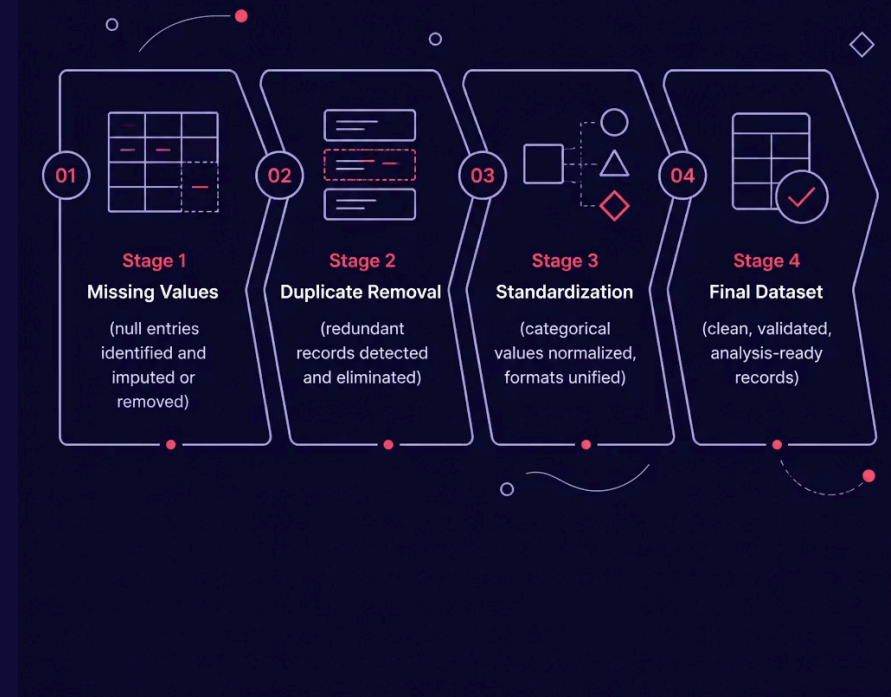
- Growing adoption of Generative AI tools in higher education
- Need to quantify impact on GPA, burnout, and skill retention
- Identify actionable trends using structured data analytics

Dataset Overview & Data Cleaning

Dataset Profile

Publicly available survey-based dataset sourced from Kaggle, capturing student demographics, AI usage patterns, academic performance, and wellbeing indicators across multiple majors and class years.

- **Key Variables:** GPA (pre/post), Weekly AI Hours, Burnout Risk, Skill Retention, Major, Year of Study
- **Source:** University student survey
- **Scope:** Multi-major, multi-year sample



50,000

Survey Responses
Total records collected

16

Key Variables
Features per record

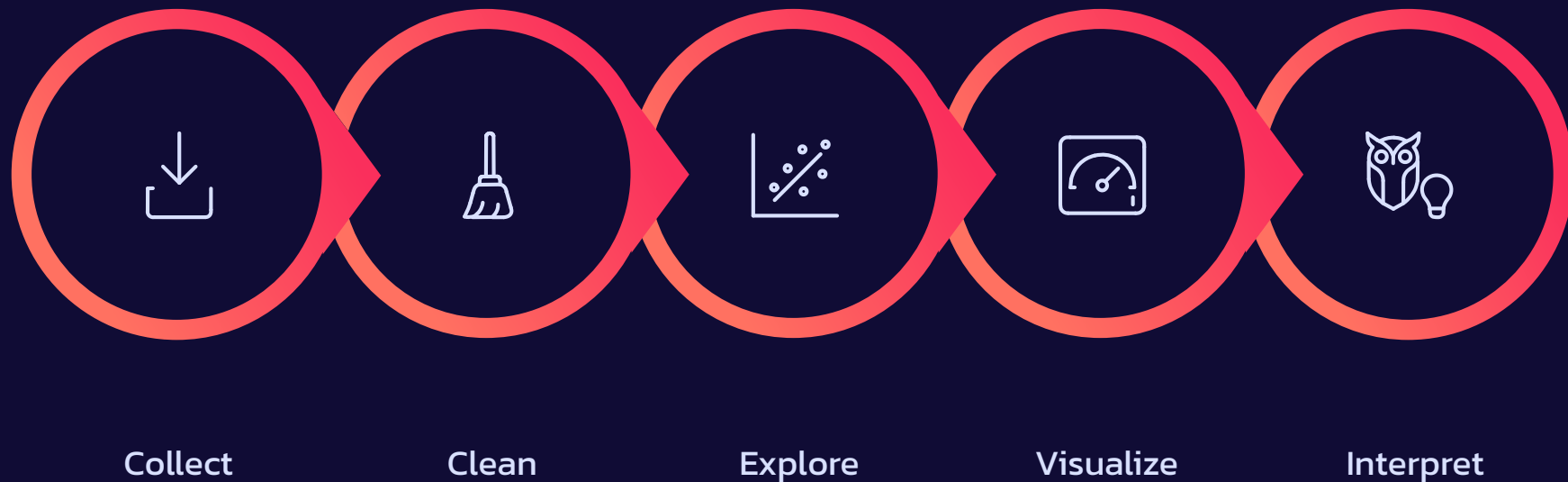
100%

Data Completeness
After cleaning

0

Duplicates
Remaining post-cleaning

Methodology & Key Metrics



The project follows a structured six-phase analytics pipeline — from raw data collection through to evidence-based recommendations — ensuring rigor and reproducibility at every stage.

3.15

Avg Pre-Semester GPA
Baseline academic performance

3.35

Avg Post-Semester GPA
Performance after AI usage

+0.20

Avg GPA Improvement
Mean semester-over-semester gain

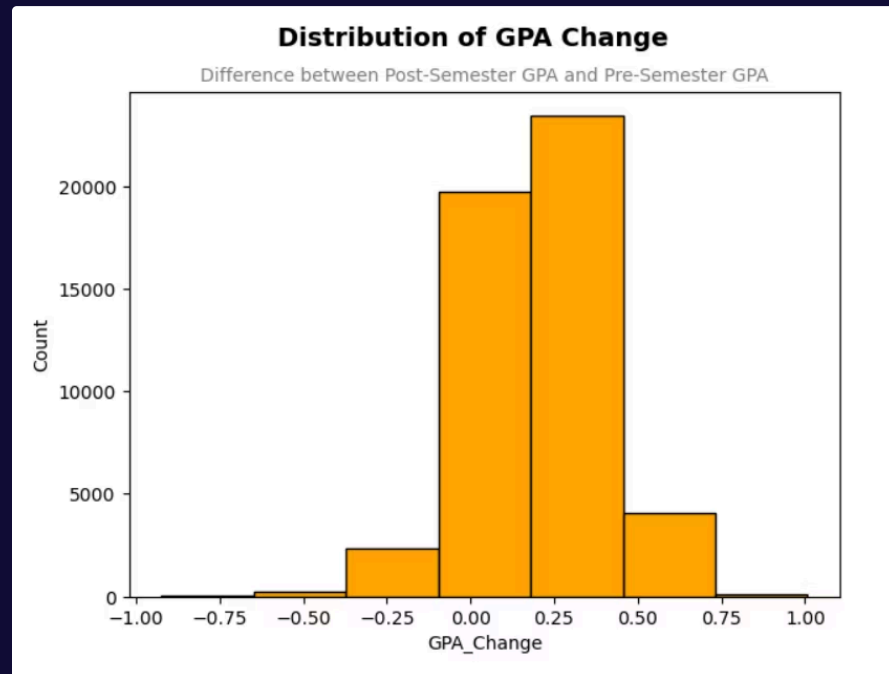
8.43

Weekly AI Hours
Average student AI usage

75.8

Skill Retention Score
Average out of 100

Exploratory Data Analysis

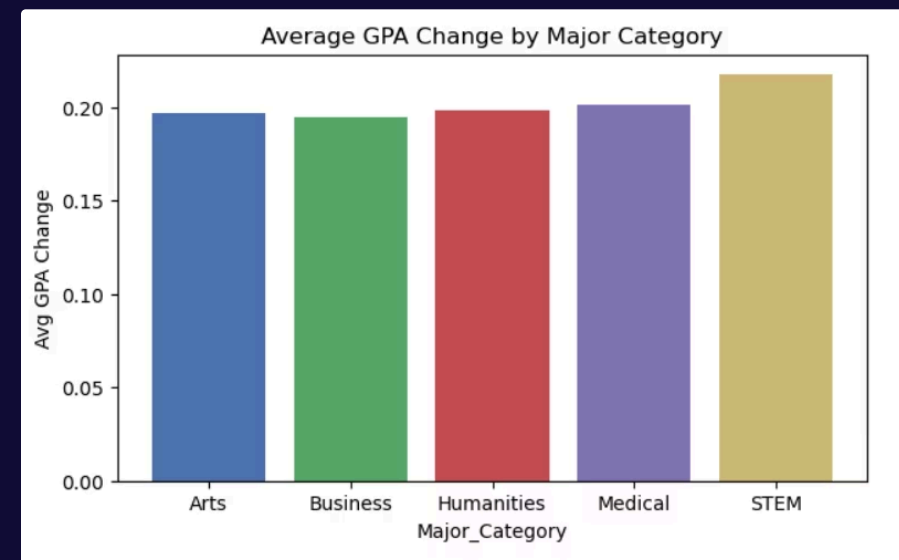


GPA Distribution

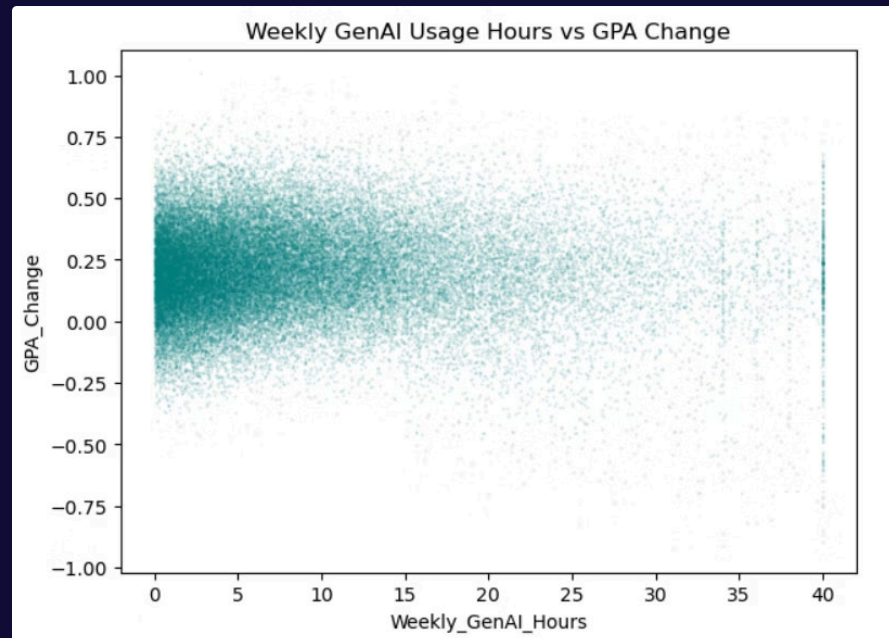
While the majority of students (87.5%) showed GPA improvement over the semester (mean change $\approx +0.20$), a notable minority (~12%) experienced a decline, with some drops as large as -0.92. This suggests GenAI usage is not uniformly beneficial, and warrants deeper analysis into which student profiles fall into the declining group.

Average GPA Change by Major Category

STEM students showed the highest average GPA improvement (+0.22) compared to other majors (range +0.19 to +0.20), a difference confirmed as statistically significant (ANOVA $p < 0.0001$) despite the modest absolute size — likely a reflection of large sample size rather than a strong practical effect. STEM students also displayed greater variance in outcomes, suggesting more divergent individual experiences with GenAI tools.

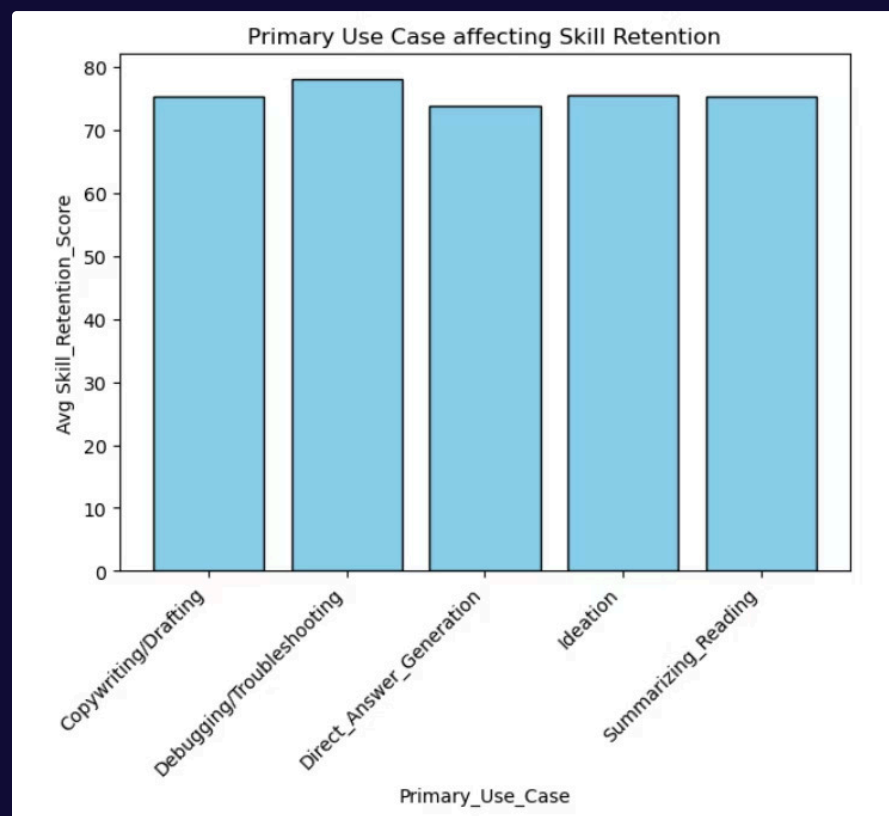


Relationship Analysis



AI Hours vs. GPA Change

Although the linear correlation between weekly GenAI usage and GPA change is negligible ($r \approx -0.05$), binning students by usage intensity reveals a non-linear relationship: GPA improvement peaks for moderate users (5–15 hrs/week, $\sim +0.23$) and declines for heavy users (35–40 hrs/week, $\sim +0.12$), suggesting diminishing or even reversing returns beyond moderate engagement.



Primary Use Case affecting Skill Retention

How students use AI matters. Using AI for **debugging and troubleshooting** is linked to the highest skill retention, while relying on **direct answers** is linked to the lowest retention and greater learning risk.

Key Takeaways

- **Debugging/Troubleshooting:** Highest skill retention (**78.1**).
- **Ideation:** Strong skill retention, supporting creative problem-solving.
- **Summarizing/Reading:** Maintains good retention by reinforcing understanding.
- **Copywriting/Drafting:** Moderate retention with consistent performance.
- **Direct Answer Generation:** Lowest skill retention (**73.7**) and highest risk of skill loss.

Key Findings & Recommendations



Moderate AI Use Boosts GPA

Best results with moderate AI use; excessive use offers limited gains.



High Usage Raises Burnout Risk

Heavy AI users show higher burnout risk.



STEM Majors Benefit Most

Academic gains are strongest in STEM disciplines.



Active Use Preserves Skills

Using AI for learning preserves critical thinking better than delegation.

→ **Encourage Balanced AI Usage**
Recommend 6–10 hrs/week.

→ **Monitor Burnout Proactively**
Provide early wellbeing support.

→ **Position AI as a Learning Assistant**
Support learning, not replacement.

→ **Build AI Literacy Programs**
Include responsible AI training in curricula.

Limitations & Future Scope

Study Limitations

Survey-Based Data

Self-reported responses may introduce recall and social desirability bias.

Correlation $\neq$ Causation

Observed relationships cannot confirm causal mechanisms without experimental design.

Sample Constraints

Limited institutional diversity may affect generalizability across student populations.

Future Research Directions

Predictive Modeling

Build ML models to forecast GPA outcomes and burnout risk from usage patterns.

Longitudinal Studies

Track the same students across multiple semesters to capture causal trends.

Student Clustering

Segment users by behavior profiles to enable personalized AI guidance.

Personalized Recommendations

Develop adaptive AI literacy pathways tailored to major and usage patterns.

Thank You

We welcome your questions, feedback, and discussion on how data analytics can guide responsible AI integration in higher education.

